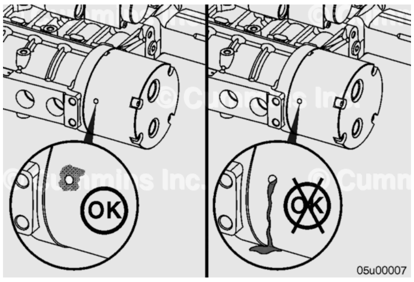
Trainings

General Information

with Electronically Actuated Injector

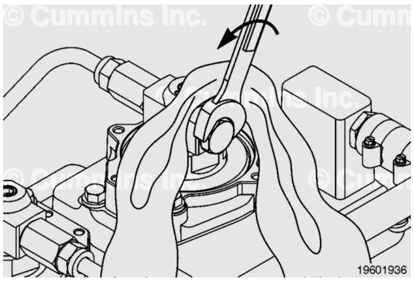
A chemical streak or buildup at the weep hole is **not** justification for fuel pump replacement. If a steady flow of oil or fuel is observed, replace the fuel pump. Contact a Cummins® Authorized Repair Location.

Note : This check can not be carried out on later pumps as the weep hole was removed.



⚠ WARNING ⚠

The pressure of the fuel in the line is sufficient to penetrate the skin and cause serious personal injury. Wear gloves and protective clothing.



Note : The pressure from the fuel system must be relieved before removing the rail pressure sensor, mechanical dump valve, or injector supply lines.

Cover the plug with a lint-free cloth. Slowly loosen the plug on the last injector of the high-pressure fuel lines 1/4 to 1/2 of a turn. The plug does **not** need to be removed to relieve the pressure.

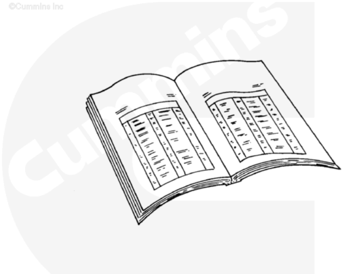
Connect INSITE™ electronic service tool and monitor the fuel rail pressure to verify the fuel pressure has bled down.

Preparatory Steps

CENTRY™

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



ck800wa

⚠ CAUTION ⚠

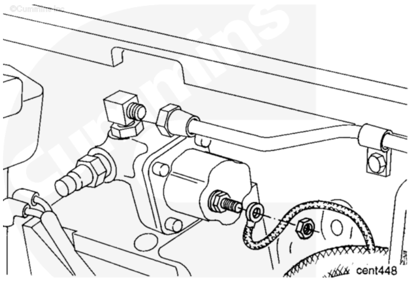
Do not over-twist or bend the CENTRY™ wiring harness. Damage to the wiring harness can occur.

- Disconnect the battery cables. Refer to the OEM service manual.
- Clean the fuel pump and the surrounding area before removing the fuel pump from the engine.
- Remove any wiring harness hold-down clamps that are within 450 mm [18 in] of the ECM. Use the following procedure from the CENTRY™ Fuel System Troubleshooting and Repair Manual, Bulletin 3666070. Refer to Procedure 019-043 in Section 19. (/qs3/pubsys2/xml/en/procedures/98/98-019-043.html)
- Remove the three ECM mounting capscrews and carefully move the ECM away from the fuel pump. Use the following procedure from the CENTRY™ Fuel System Troubleshooting and Repair Manual, Bulletin 3666070. Refer to Procedure 019-031 in Section 19. (/qs3/pubsys2/xml/en/procedures/98/98-019-031.html)
- Disconnect the electronic fuel control electrical wires from the electronic fuel control valve. Use the following procedure from the CENTRY™ Fuel System Troubleshooting and Repair Manual, Bulletin 3666070. Refer to Procedure 019-030 in Section 19. (/qs3/pubsys2/xml/en/procedures/98/98-019-030.html)
- Disconnect the wires to the fuel shutoff valve. Use the following procedure from the CENTRY™ Fuel System Troubleshooting and Repair Manual, Bulletin 3666070. Refer to Procedure 019-050 in Section 19. (/qs3/pubsys2/xml/en/procedures/98/98-019-050.html)

PT

⚠ WARNING ⚠

Fuel is flammable. Keep all cigarettes, flames, pilot lights, arcing equipment, and switches out of work area and areas sharing ventilation to reduce the possibility of severe personal injury or death when working on the fuel system.



Disconnect the fuel hose, tubing, and the following:

- Fuel inlet supply (**not** shown)
- Gear pump cooling drain (**not** shown)
- Fuel rail pressure line
- Fuel pressure sensing line to the STC valve, if equipped.

with Electronically Actuated Injector

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⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

A very small amount of dirt and debris can be very harmful to the fuel pump. Extra care is required to keep the fuel connections clean during removal and installation. Connections must be covered immediately to keep them clean when components are removed from the fuel pump.

- Disconnect the battery cables. Refer to the OEM service manual.
- Clean the fuel pump and the surrounding area and dry with compressed air.
- Remove the fuel drain line from the fuel pump. Refer to Procedure 006-013 in Section 6. (/qs3/pubsys2/xml/en/procedures/28/28-006-013.html)
- Remove the injector supply line from the fuel pump. Refer to Procedure 006-051 in Section 6. (/qs3/pubsys2/xml/en/procedures/28/28-006-051-tr.html)
- Remove the fuel supply lines from the fuel pump. Use a clean plastic plug to protect the fuel supply ports. Refer to Procedure 006-024 in Section 6. (/qs3/pubsys2/xml/en/procedures/28/28-006-024-tr.html)
- Remove the intake manifold pressure sensors and intake manifold air temperature sensor that are located above the fuel pump. Use the following two procedures in the Troubleshooting and Repair Manual, Electronic Control System, QSK50 and QSK60 Modular Common Rail System Series Engine, Bulletin 4021533. Refer to Procedure 019-061 in Section 19. (/qs3/pubsys2/xml/en/procedures/05/05-019-061.html) Refer to Procedure 019-059 in Section 19. (/qs3/pubsys2/xml/en/procedures/05/05-019-059.html)

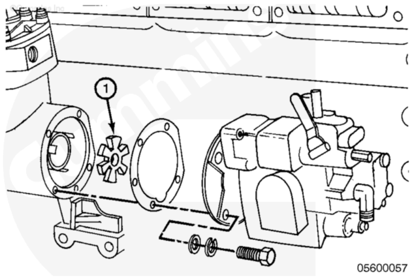
Remove

with Mechanically Actuated Injector

Remove the four mounting capscrews and the fuel pump.

Remove the drive coupling.

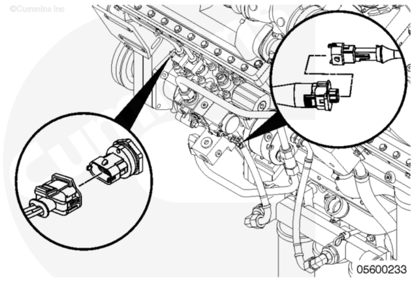
Remove and discard the gasket.



with Electronically Actuated Injector

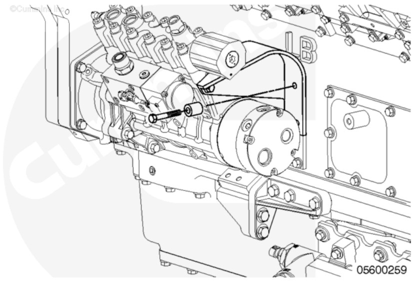
Note : Shown throughout the procedure is the fuel pump for the QSK50 engine. Although there is a difference in the number of cylinders in the pump, the procedure remains the same.

Disconnect the wiring harness to the fuel pump pressurizing assembly and injector metering rail 1 pressure sensor.



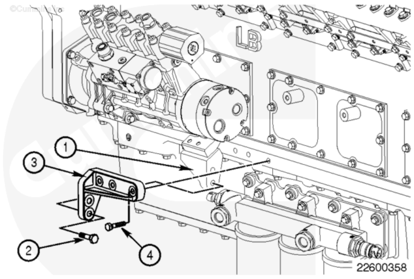
Remove the two capscrews and spacers securing the upper fuel support bracket to the hand hole cover.

Do **not** remove the upper support bracket from the fuel pump at this time.

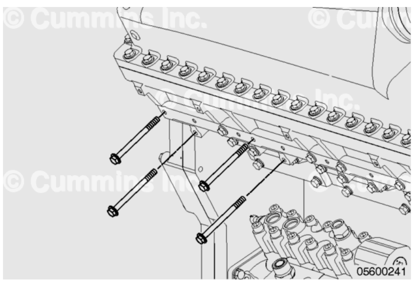


Remove the two capscrews and washers (2) securing the pump support brace (3) to the support bracket (1) attached to the pump.

Remove the three capscrews and washers (4) securing the pump support brace to the engine block.

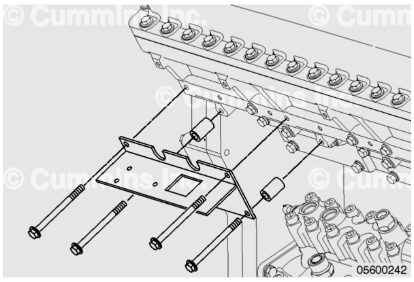


Remove the four intake manifold capscrews.



⚠ CAUTION ⚠

The intake manifold pressure and temperature sensors must be removed to avoid damage prior to installing the mounting bracket.

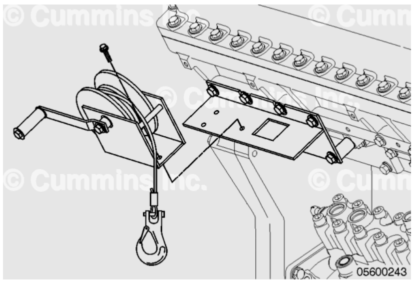


Align the mounting bracket from the fuel pump lifting fixture kit, Part Number 5299015, or equivalent, with the cylinder block.

Install the two spacers and capscrews. Tighten the capscrews.

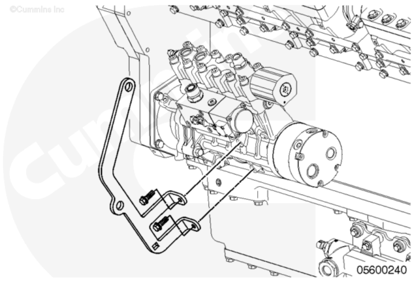
Torque Value: 68 n•m [50 ft-lb]

Place the winch onto the mounting bracket. Install the three capscrews, lock washers, and nuts. Tighten the capscrews.



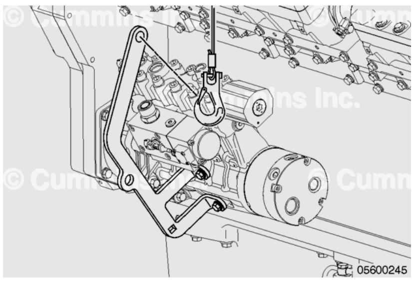
Torque Value: 68 n•m [50 ft-lb]

Install the lifting bracket on the fuel pump. Use the two capscrews.



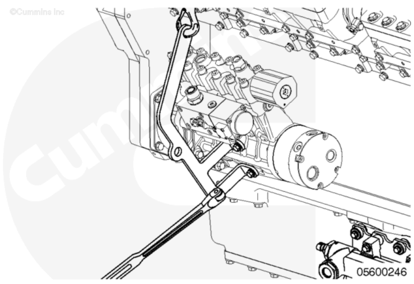
Torque Value: 45 n•m [33 ft-lb]

Attach the lifting hook to the lifting bracket.



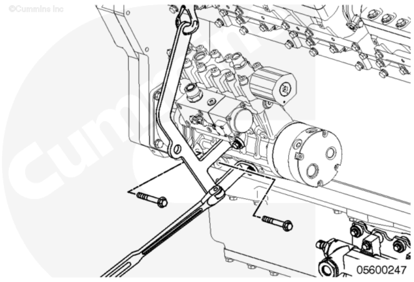
Turn the winch handle until the lifting cable is tight.

Insert a 1/2-inch breaker bar into the square notch on the lifting bracket. This will aid in the removal, installation, and maneuvering of the fuel pump.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



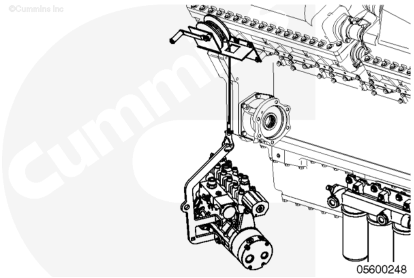
Remove the six capscrews securing the fuel pump to the fuel pump drive.

Use the 1/2-inch breaker bar to negotiate the fuel pump out of the gear housing. Remove the fuel pump from the engine.

Lower the fuel pump using the hand winch.

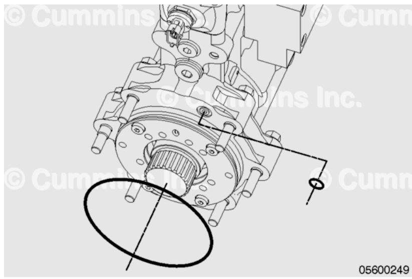
Remove the hook from the lifting bracket.

Remove the 1/2-inch breaker bar from the fuel pump lifting bracket.



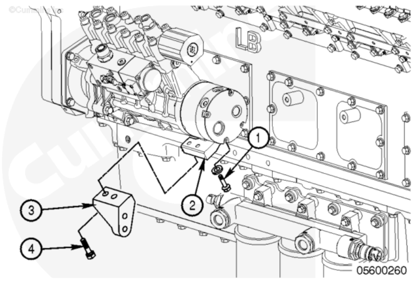
Remove and discard the o-ring that is used to seal the oil drilling from the fuel pump drive to the fuel pump.

Remove and discard the o-ring that is used to seal the fuel pump to the fuel pump drive.



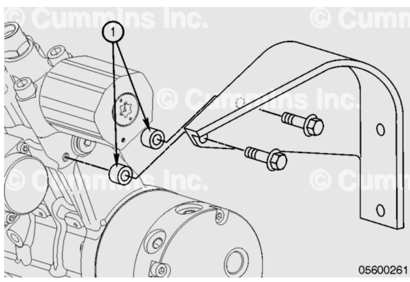
Remove the two capscrews and washers (4) securing the support brace (3) to the fuel pump support bracket (2) attached to the fuel pump.

Remove the two capscrews (1) and fuel pump support bracket (2) attached to the fuel pump.



Remove the two capscrews and spacers (1) securing the upper fuel pump support bracket to the fuel pump.

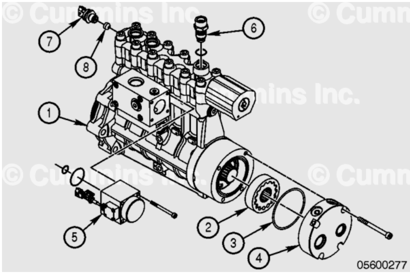
Note : The spacers must be retained for installation. The upper support bracket must not be installed without the spacers.



Disassemble

with Electronically Actuated Injector

Note : The fuel pump is manufactured using metric fasteners. Use metric size tools during disassembly of the pump.



Remove the mechanical dump valve (6) from the fuel pump (1). Refer to Procedure 006-061 in Section 6 (/qs3/pubsys2/xml/en/procedures/102/102-006-061.html).

Remove the injector metering rail 1 pressure sensor (7).

Remove the sealing washer (8). Use a magnet, then discard the sealing washer.

Note : Older versions of the fuel pump did not use a sealing washer.

Plug the injector metering rail 1 pressure sensor port with a clean plastic plug.

Remove the fuel pump pressurizing assembly (5). Refer to Procedure 005-232 in Section 5 (/qs3/pubsys2/xml/en/procedures/102/102-005-232.html).

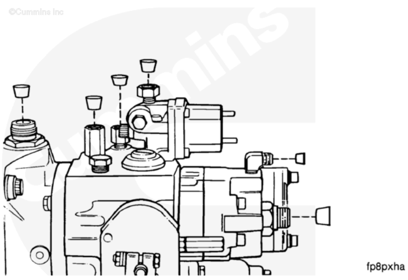
Remove the four capscrews and gerotor cover plate (4). Discard the o-ring (3).

Remove the gerotor gear (2).

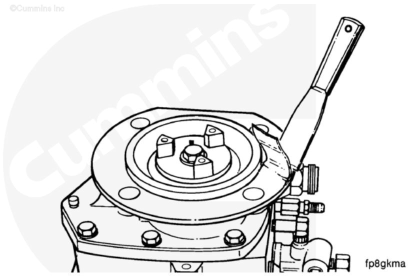
Clean and Inspect for Reuse

with Mechanically Actuated Injector

Install plastic cup plugs or tape on all of the openings of the fuel pump to prevent dirt or cleaning solvent from entering the fuel pump.

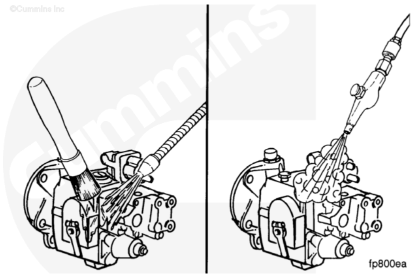


Remove the gasket material from the front cover gasket sealing surface.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

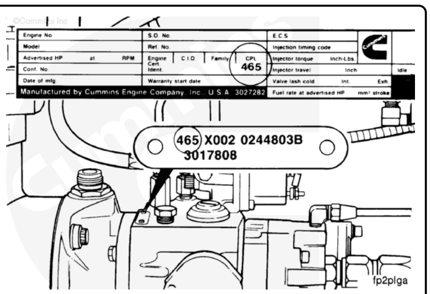
Use a cleaning solvent that will not harm aluminum.

Use a brush and solvent to clean the fuel pump exterior.

Dry with compressed air.

Check to make sure the same Control Parts List (CPL) number is on both the fuel pump dataplate and the engine dataplate.

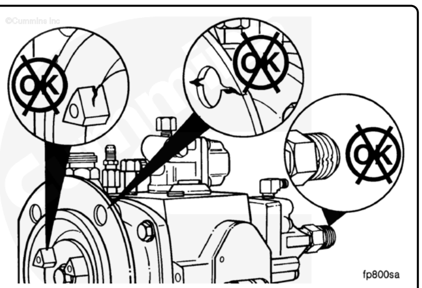
If the CPL number on the fuel pump dataplate does **not** match the CPL number on the engine dataplate, the fuel pump **must** be calibrated again to the correct specification for the engine.



Check the fuel pump body and front cover for cracks or other damage.

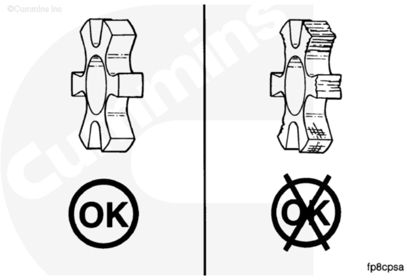
Check the fuel pump assembly for damaged capscrews and damaged or loose fuel fittings.

Check the drive coupling lugs for excessive wear or other damage.

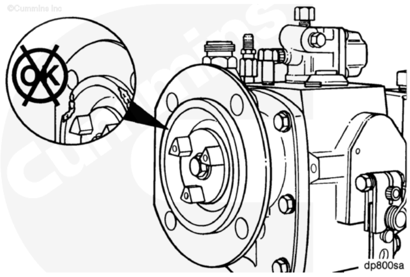


Note : All K38 and K50 engines must have the white or light green nylon coupling spider.

Check the drive coupling spider for cracks or other damage.



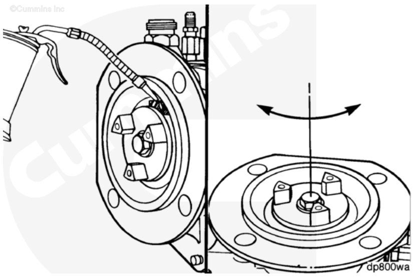
Check the front cover pilot to make sure the pilot is continuous, without cracks or gaps.



Use clean 15W-40 oil to lubricate the tachometer drive gear,

Turn the fuel pump shaft by hand. The shaft **must** rotate freely.

If the shaft does **not** rotate freely, the pump **must** be disassembled for further inspection. See the PT™ Fuel Pump (Type G) Rebuild and Calibration Instructions, Bulletin 3379084.

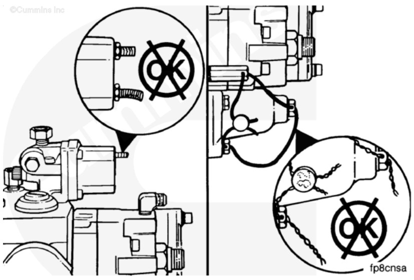


Check the fuel pump shutoff valve for loose or damaged electrical terminals.

Check the fuel pump tamper proof seals.

If the tamper proof seals have been broken or are missing, the fuel pump **must** be calibrated. See the PT™ Fuel Pump (Type G) Rebuild and Calibration Instructions, Bulletin 3379084.

Install new tamper proof seals.

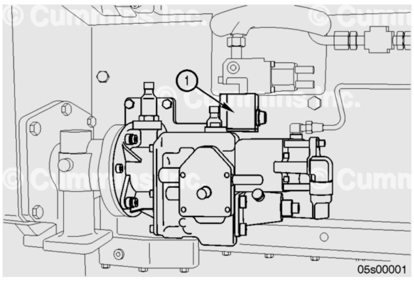


Check the fuel pump shutoff valve block (1) filter screen for signs of debris and dirt, if fitted.

Note : The valve block may be either mounted on the fuel pump, as illustrated, or remote mounted depending on the engine configuration.

Remove the filter cap and spring.

Remove the filter screen and clean any signs of debris. If the filter screen is damaged, it **must** be replaced.

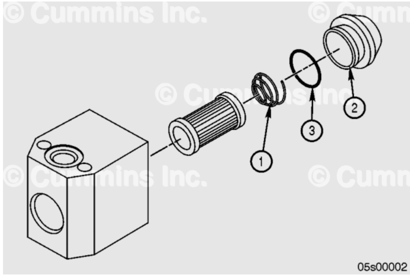


Install the filter screen (4) into the block (5). The filter must be inserted with the opening facing the inside of the block and the solid end facing the cap.

Install the spring (1). The spring must be inserted with the smaller radius facing the filter screen.

Make sure the o-ring (3) is fitted to the cap (2).

Install the cap and tighten.



Torque Value: 11 to 16 n•m [97 to 142 in-lb]

with Electronically Actuated Injector

⚠ WARNING ⚠

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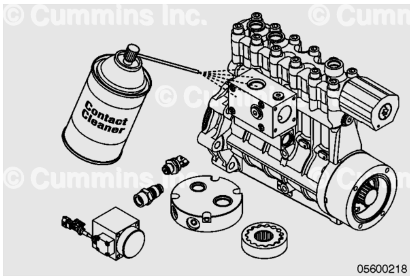
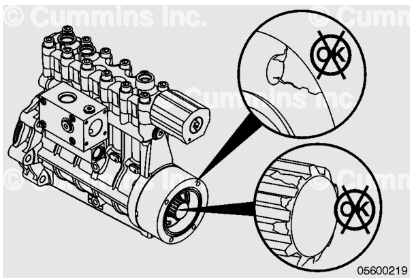


Clean all parts with electrical contact cleaner, Part Number 3824510, or equivalent.

Inspect the main pump for cracks, leaks, drive shaft damage, and any other damage.

Replace the pump if damage is found.



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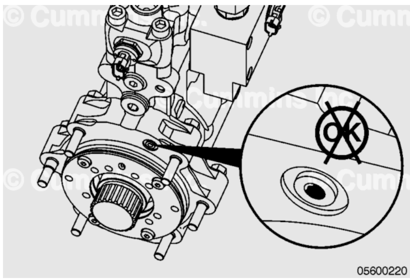


⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

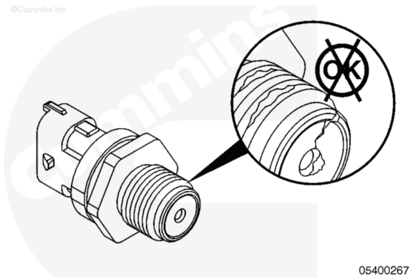
Inspect the oil supply and return passages for restrictions, dirt, and debris.

Note : Some pumps have seals installed in the two lower oil drain passages. These should not be removed.

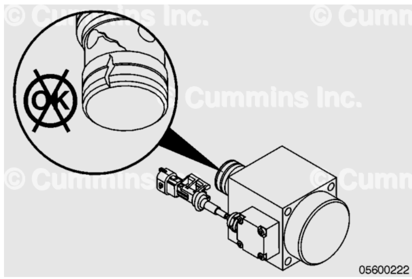
Clean the passages with electrical contact cleaner, Part Number 3824510, or equivalent.

Dry with compressed air.

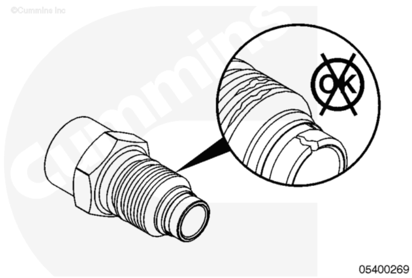
Inspect the pressure sensor for cracks, stripped threads, or other damage. Use the following procedure in the Troubleshooting and Repair Manual Electronic Control System, QSK50 and QSK60 Modular Common Rail System Series Engines, Bulletin 4021533. Refer to Procedure 019-115 in Section 19. (/qs3/pubsys2/xml/en/procedures/102/102-019-115.html)



Inspect the fuel pump pressurizing assembly for damage. Refer to Procedure 005-232 in Section 5. (/qs3/pubsys2/xml/en/procedures/102/102-005-232.html)

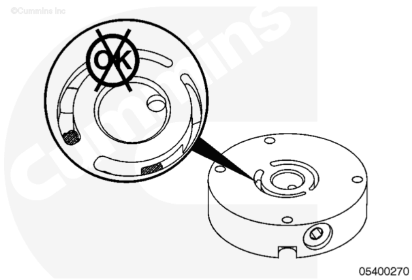


Inspect the mechanical dump valve. Refer to Procedure 006-061 in Section 6. (/qs3/pubsys2/xml/en/procedures/102/102-006-061.html)

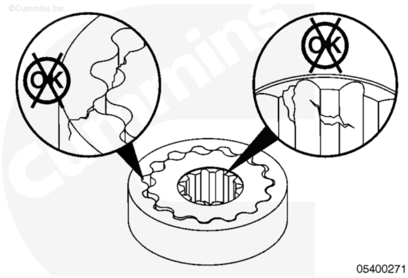


Inspect the cover plate for cracks and other damage. The cover plate also contains the regulator for the fuel pump.

Check the regulator for restriction passage.



Note : The pump gears are a matched set. If the pump gears are to be replaced, they must be replaced as a set, not individual gears.

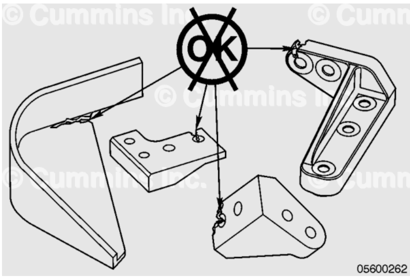


Inspect the pump gears for wear, cracks, debris, and broken or missing teeth.

Check for torsional play in the inner gear to the drive spline in the pump.

Replace the pump gears if damage is found.

Inspect the four fuel pump support brackets for corrosion, cracks, wear, and any other damage.

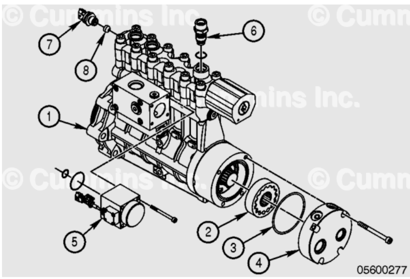


Replace if damage is found.

Assemble

with Electronically Actuated Injector

Install the gerotor gear (2) into the fuel pump (1).



Lubricate the new o-ring (3) with clean engine oil and install the o-ring onto the gerotor cover plate (4).

Install the gerotor cover plate (4) and the four capscrews.

Tighten the capscrews.

Torque Value: 19 n•m [168 in-lb]

Install the fuel pump pressurizing assembly (5). Refer to Procedure 005-232 in Section 5. (/qs3/pubsys2/xml/en/procedures/102/102-005-232.html)

Tighten the capscrews.

Note : The sealing washer (8) is a single use seal and must be replaced when the injector metering rail pressure sensor (7) is removed or loosened.

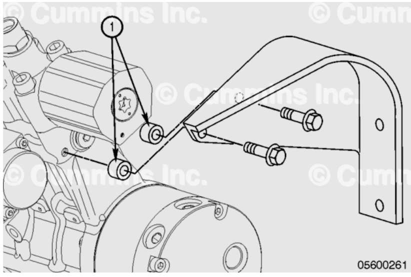
Install the new sealing washer (8), then install the injector metering rail 1 pressure sensor (7).

Torque Value: 136 n•m [100 ft-lb]

Note : Older versions of the fuel pump did not use a sealing washer. The use of a sealing washer on an older version of the fuel pump will not allow proper thread engagement of the pressure sensor.

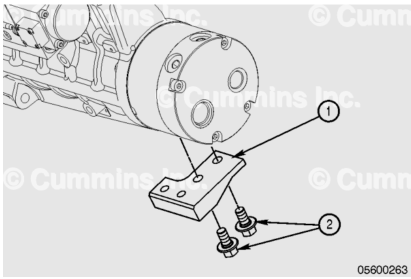
Install the mechanical dump valve (6). Refer to Procedure 006-061 in Section 6. (/qs3/pubsys2/xml/en/procedures/102/102-006-061.html)

Install the upper fuel pump support bracket to the fuel pump using the two capscrews and spacers (1). Check that both spacers are used. Hand-tighten the capscrews at this time. Do **not** tighten.



Install the two capscrews (2) and fuel pump support bracket (1) attached to the fuel pump.

Torque Value: 113 n•m [83 ft-lb]

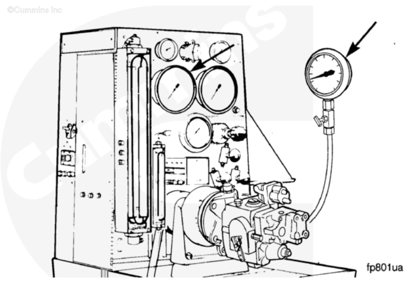


Calibrate

with Mechanically Actuated Injector

Note : Calibration of the fuel pump requires special equipment and must be done at a Cummins® Authorized Repair Location.

Calibrate the fuel pump.



with Electronically Actuated Injector

Calibration is **not** required on engines equipped with modular common rail fuel systems.

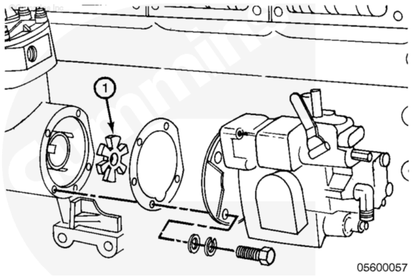
Install

with Mechanically Actuated Injector

Install the drive coupling (1), new gasket, fuel pump, and four capscrews.

Tighten the capscrews.

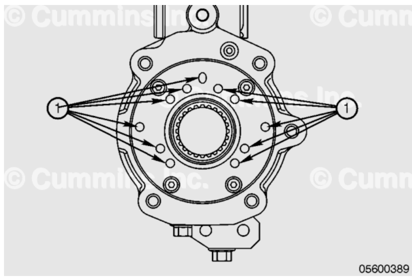
Torque Value: 48 n•m [35 ft-lb]



with Electronically Actuated Injector

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



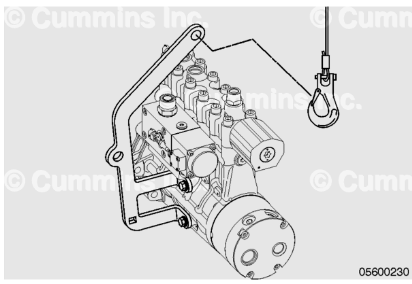
On a flat level surface, place the pump on its end with the gerotor-side down.

Pour 104 ml [3.5 oz] of fresh, clean lubricating oil into one of the eleven available face drillings (1).

Attach the lifting bracket to the fuel pump using the supplied hardware.

Torque Value: 45 n•m [33 ft-lb]

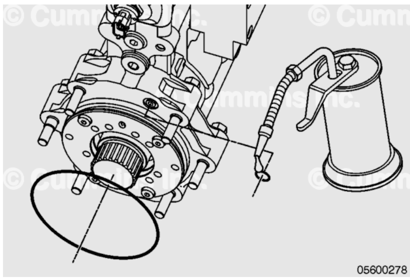
Attach the lifting hook from the winch to the lifting bracket.



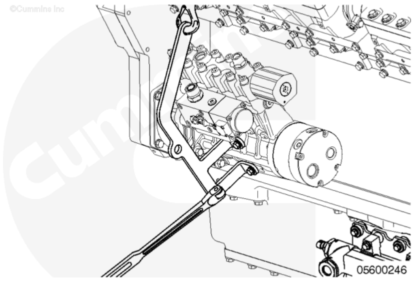
Use clean engine oil to lubricate both new o-rings.

Install the smaller o-ring that is used to seal the oil drilling from the fuel pump drive to the fuel pump.

Install the large o-ring that is used to seal the fuel pump drive to the fuel pump.

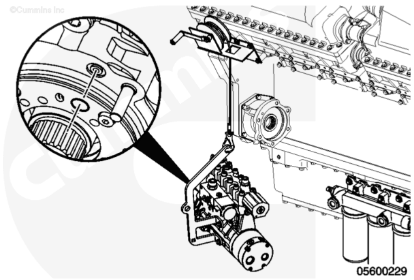


Insert a ½ inch breaker bar or ratchet into the square notch on the lifting bracket.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



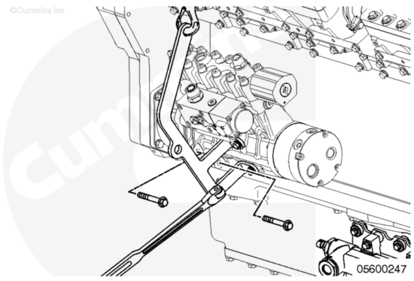
⚠ CAUTION ⚠

Verify that the o-ring on the flange face of the fuel pump is in place. Replace the o-ring if it is missing. Failure to do so will cause failure of the fuel pump due to lack of lubrication.

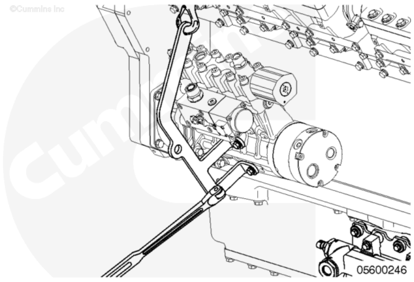
Use the hand winch and raise the fuel pump into place.

Install the fuel pump into the fuel pump drive. Use the breaker bar or ratchet to help guide the fuel pump into place.

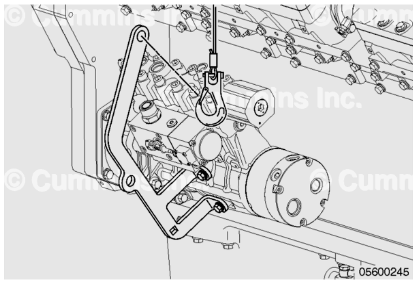
Rotate the pump to align the mounting holes and install the six capscrews. Hand tighten the capscrews at this time.



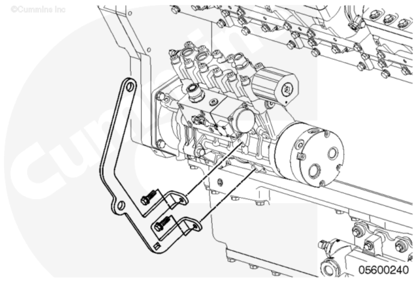
Remove the ½ inch breaker bar or ratchet from the lifting bracket.



Remove the hook from the lifting bracket.

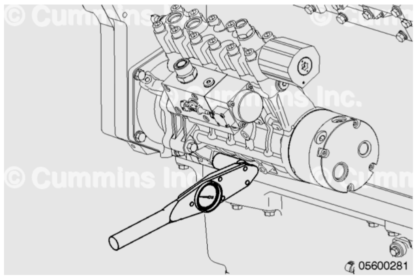


Remove the capscrews and lifting bracket from the fuel pump.

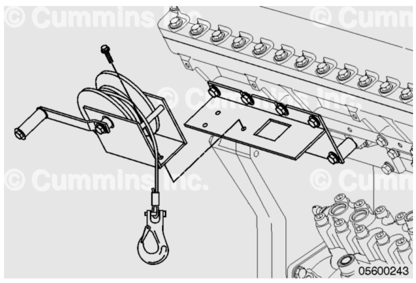


Tighten the six fuel pump mounting capscrews.

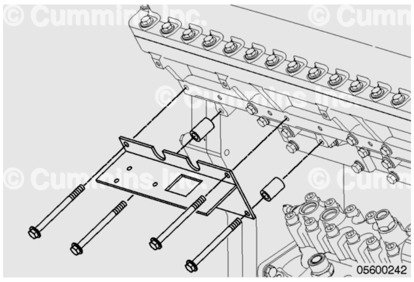
Torque Value:
Fuel Pump Mounting Capscrews 117 n•m [86 ft-lb]



Remove the three capscrews, locking washers, and nuts that are securing the winch to the support bracket.

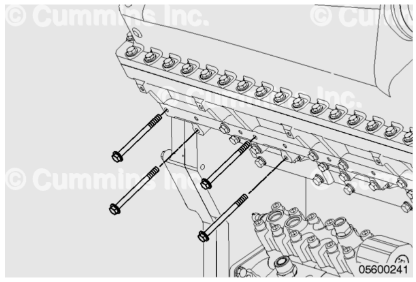


Remove the four capscrews and spacers that are securing the winch support bracket to the intake manifold.



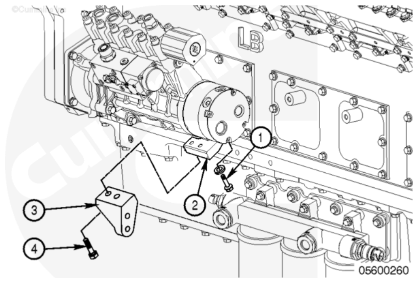
Install the intake manifold capscrews.

Torque Value: 58 n•m [43 ft-lb]



Install the two capscrews and washers (4) securing the fuel pump support bracket (3) to the bracket (2) attached to the fuel pump.

Only hand-tighten the capscrews at this time.



Install the three capscrews and washers (4) securing the fuel pump support bracket (3) to the engine block.

Only hand-tighten the capscrews at this time.

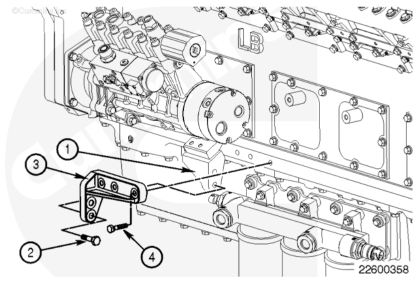
Install the two capscrews and washers (2) securing the fuel pump support bracket (3) to the fuel pump support bracket (1).

Only hand-tighten the capscrews at this time.

Once all the capscrews are installed and hand tightened, tighten the capscrews.

Torque Value:
Bracket to Bracket Capscrews (2) 45 n•m [33 ft-lb]

Torque Value:
Bracket to Engine Block Capscrews (4) 45 n•m [33 ft-lb]



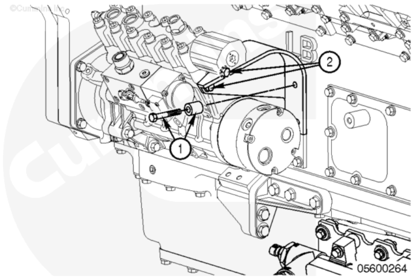
Install the two capscrews and spacers (1) that attach the upper fuel support bracket to the hand hole cover.

Tighten the capscrews.

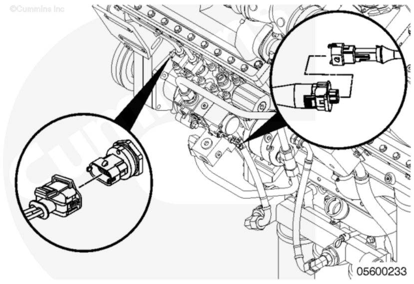
Torque Value: 45 n•m [33 ft-lb]

Tighten the two capscrews (2) that attach the bracket to the fuel pump.

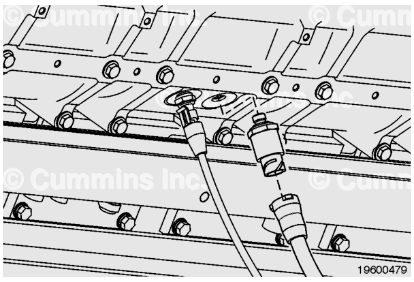
Torque Value: 45 n•m [33 ft-lb]



Connect the wiring harness to the fuel pump pressurizing assembly and injector metering rail 1 pressure sensor.



Install the intake manifold pressure sensors and intake manifold air temperature sensor that are located above the fuel pump. Use the following procedures in the QSK50 and QSK60 Modular Common Rail System Series Engine Electronic Control System Troubleshooting and Repair Manual, Bulletin 4021533.



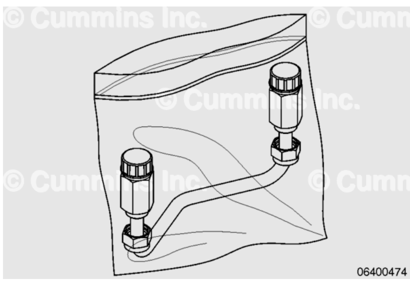
- Refer to Procedure 019-061 in Section 19. (/qs3/pubsys2/xml/en/procedures/05/05-019-061.html)
- Refer to Procedure 019-059 in Section 19. (/qs3/pubsys2/xml/en/procedures/05/05-019-059.html)

Flush

with Electronically Actuated Injector

Follow these flush steps anytime debris which may contaminate the fuel syste, is believed to have fallen into the fuel pump.

A spare U-shaped injector supply line and a container to catch 3.78 L [1 gal], or more, fuel are required.

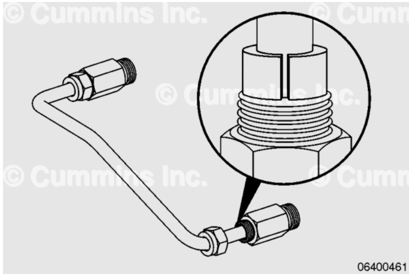


Remove the injector supply line from the clean bag.

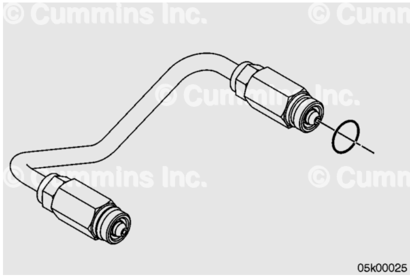
Lubricate the pre-sliced grommets with clean engine oil on the outside and inside diameters.

Install the grommets on the outer wall of the injector supply line.

Remove the yellow threaded caps from the ends of the injector supply line and discard the caps.



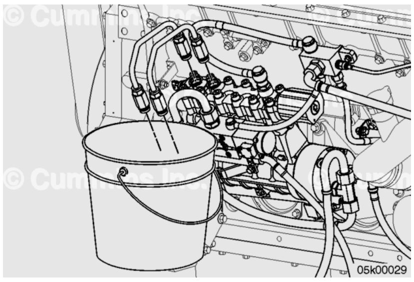
Install a new o-ring on one cone end of the injector supply line.



Install the two U-shaped injector supply lines into the fuel pump outlets.

Start the first thread of the connector nuts by hand.

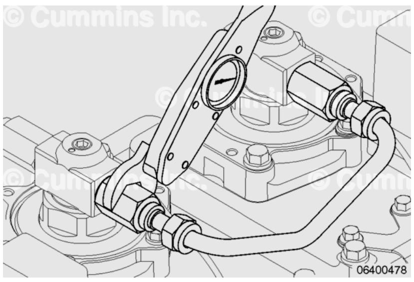
Route the injector supply lines into a suitable container that can hold 3.78 L [1 gal], or more, of fuel.



Note : Although the illustration does not show tightening the fuel line on the fuel pump, the techniqe is the same.

Tighten the connector nut with a suitable M27 [1-1/16 in] crow's foot and torque wrench.

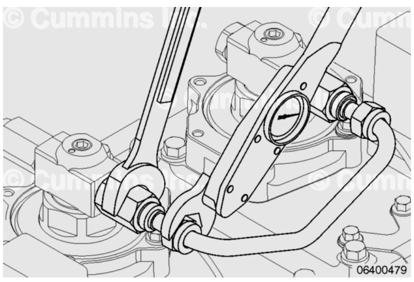
Torque Value: 45 n•m [33 ft-lb]



Note : Although the illustration does not show tightening the fuel line on the fuel pump, the techniqe is the same.

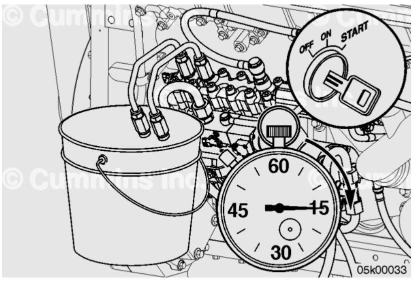
Install the rectangular grommet nuts onto the connector nuts with a suitable M24 [15/16 in] crow's foot and torque wrench. Support the connector nut with a wrench while tightening the rectangular grommet nut.

Torque Value: 45 n•m [33 ft-lb]



⚠ WARNING ⚠

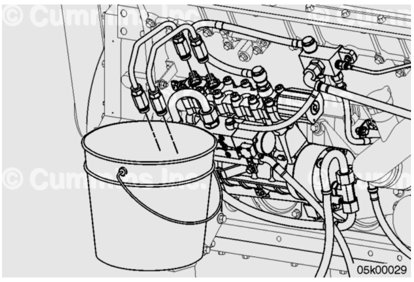
Depending on the circumstance, diesel fuel is flammable. When inspecting or performing service or repairs on the fuel system, to reduce the possibility of fire and resulting severe personal injury, death or property damage, never smoke or allow sparks or flames (such as pilot lights, electrical switches, or welding equipment) in the work area.



Confirm the injector supply lines are routed into the container.

Crank the engine for 60 seconds in 15 second increments to flush fuel through the pump.

Loosen the rectangular grommet nuts on the U-shaped injector supply lines.



Loosen the connector nuts and remove the injector supply lines from the fuel pump.

Remove and discard the o-rings.

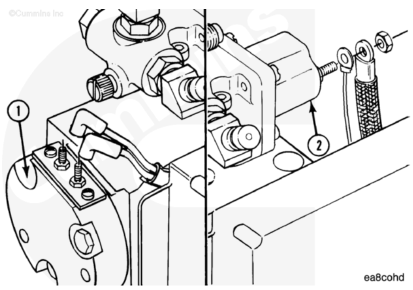
Immediately after removing the injector supply lines, cap the ends with new yellow threaded caps to prevent debris from entering the lines.

Finishing Steps

CENTRY™

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



⚠ CAUTION ⚠

Do not over twist or bend the CENTRY™ wiring harness. Damage to the wiring harness can occur.

- Connect the wires to the fuel shutoff valve. Use the following procedure from the CENTRY™ Fuel System Troubleshooting and Repair Manual, Bulletin 3666070.

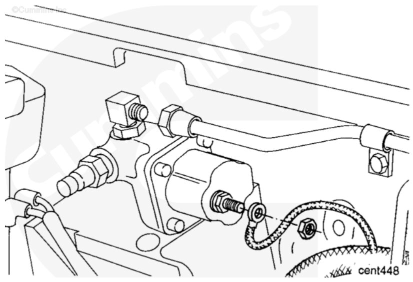
Refer to Procedure 019-050 in Section 19.
(/qs3/pubsys2/xml/en/procedures/98/98-019-050.html)

- Connect the electronic fuel control electrical wires to the electronic fuel control valve. Use the following procedure from the CENTRY™ Fuel System Troubleshooting and Repair Manual, Bulletin 3666070. Refer to Procedure 019-030 in Section 19.
(/qs3/pubsys2/xml/en/procedures/98/98-019-030.html)
- Install the three ECM mounting capscrews and carefully move the ECM away from the fuel pump. Use the following procedure from the CENTRY™ Fuel System Troubleshooting and Repair Manual, Bulletin 3666070. Refer to Procedure 019-031 in Section 19.
(/qs3/pubsys2/xml/en/procedures/98/98-019-031.html)
- Install any wiring harness hold-down clamps that are within 450 mm [18 in] of the ECM. Use the following procedure from the CENTRY™ Fuel System Troubleshooting and Repair Manual, Bulletin 3666070. Refer to Procedure 019-043 in Section 19.
(/qs3/pubsys2/xml/en/procedures/98/98-019-043.html)
- Connect the battery cables. Refer to the OEM service manual.
- Operate the engine and check for leaks.

PT

Connect the fuel hose, tubing, and the following:

- Fuel inlet supply (**not** shown)
- Gear pump cooling drain (**not** shown)
- Fuel rail pressure line
- Fuel pressure sensing line to STC valve, if equipped.



with Electronically Actuated Injector

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

Fuel is flammable. Keep all cigarettes, flames, pilot lights, arcing equipment, and switches out of work area and areas sharing ventilation to reduce the possibility of severe personal injury or death when working on the fuel system.

- Install the fuel supply line to the fuel pump. Refer to Procedure 006-024 in Section 6. (/qs3/pubsys2/xml/en/procedures/28/28-006-024-tr.html)
- Install the injector supply line to the fuel pump. Refer to Procedure 006-051 in Section 6. (/qs3/pubsys2/xml/en/procedures/28/28-006-051-tr.html)
- Connect the fuel return line to the fuel pump. Refer to Procedure 006-013 in Section 6. (/qs3/pubsys2/xml/en/procedures/28/28-006-013.html)
- Connect the battery cables. Refer to the OEM service manual.
- Operate the engine and check for leaks.

Last Modified: 12-Aug-2020